GNS3 VLAN & Inter-VLAN Routing Lab

Overview

This document provides a comprehensive summary of a GNS3 network simulation lab demonstrating advanced switching and routing concepts. The lab focuses on VLAN configuration, inter-VLAN routing, dynamic IP address assignment (DHCP), and multi-subnet network connectivity across two separate network segments connected through routers and switches. This lab validates practical implementation of enterprise network architecture patterns used in real-world deployments.

Network Topology

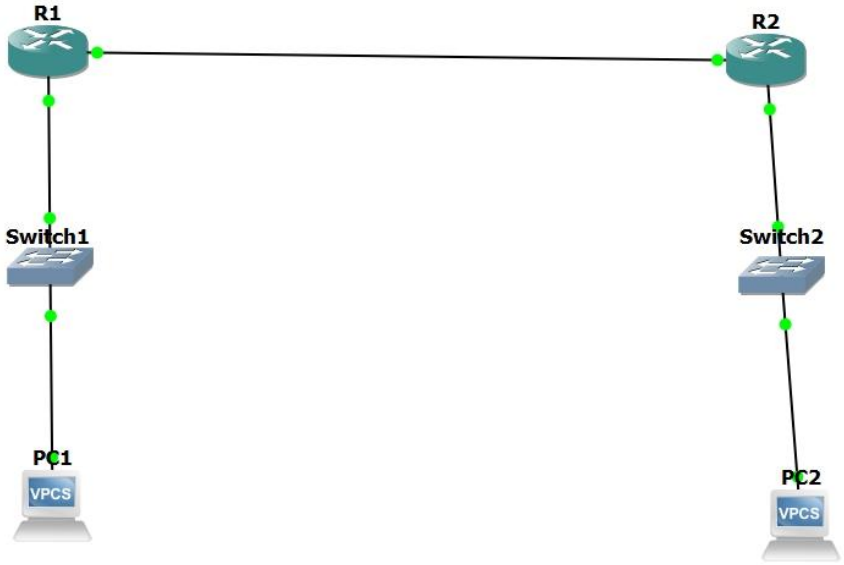

Figure 1: Figure 1: Complete network topology showing R1 and R2 routers, Switch1 and Switch2, with PC1 in the left network segment and PC2 in the right network segment

Network Architecture

Dual Segment Design

The lab implements a geographically distributed network with two distinct subnets:

- **Left Network Segment**: Connected via Switch1 to R1

  - Switch1 manages local area connectivity
  - PC1 is connected to Switch1
  - Network operates on 192.168.1.0/24 subnet
- **Right Network Segment**: Connected via Switch2 to R2
  - Switch2 manages local area connectivity
  - PC2 is connected to Switch2
  - Network operates on 192.168.2.0/24 subnet
- **Router Interconnection**: R1 and R2 are directly connected via back-to-back link
  - R1 FastEthernet0/0: 1.1.1.1/24
  - R2 FastEthernet0/0: 1.1.1.2/24
  - Enables inter-network communication and routing

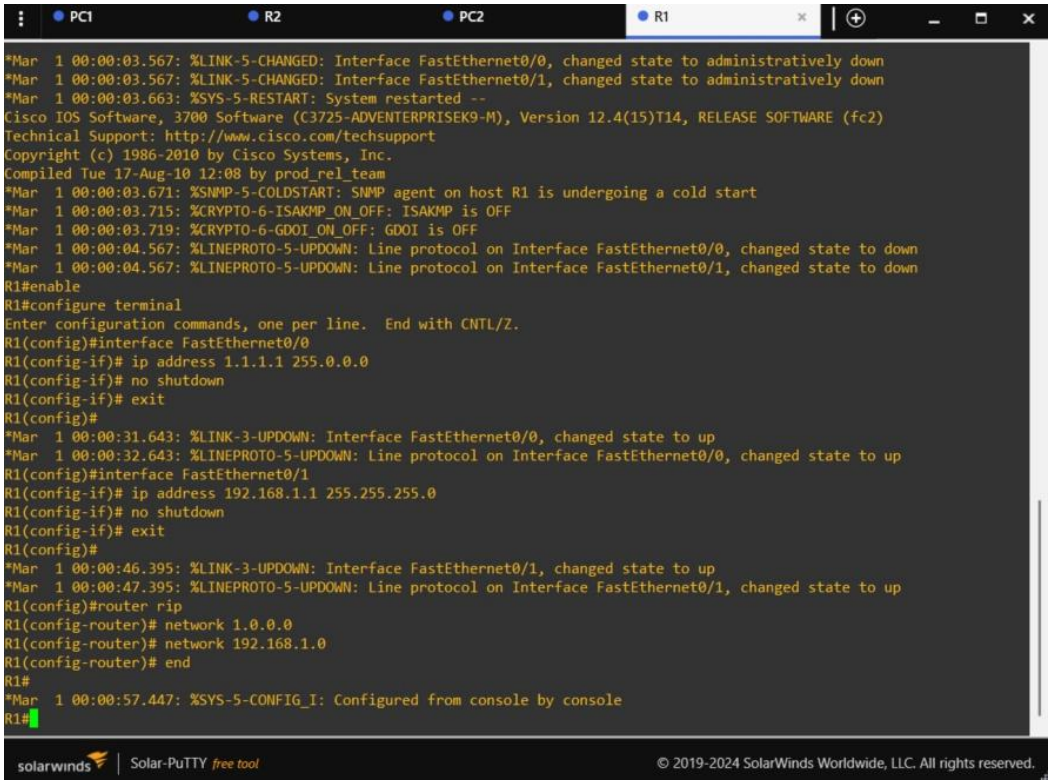

Figure 2: Figure 2: R1 router configuration showing FastEthernet0/0 interface (1.1.1.1/24), FastEthernet0/1 interface (192.168.1.1/24), and routing protocol setup for inter-network communication

Router Configuration Details

R1 Configuration

R1 serves as the gateway for the left network segment (192.168.1.0/24):

- **FastEthernet0/0**: 1.1.1.1 255.0.0.0 (backbone link to R2)
- **FastEthernet0/1**: 192.168.1.1 255.255.255.0 (local network gateway)
- **Routing Protocol**: OSPF network configuration with 192.168.1.0 and 1.0.0.0 networks
- **Status**: Both interfaces operational and routing enabled

R2 Configuration

R2 serves as the gateway for the right network segment (192.168.2.0/24):

- **FastEthernet0/0**: 1.1.1.2 255.0.0.0 (backbone link to R1)
- **FastEthernet0/1**: 192.168.2.1 255.255.255.0 (local network gateway)
- **Routing Protocol**: OSPF network configuration with 192.168.2.0 and 1.0.0.0 networks
- **Status**: Both interfaces operational and routing enabled

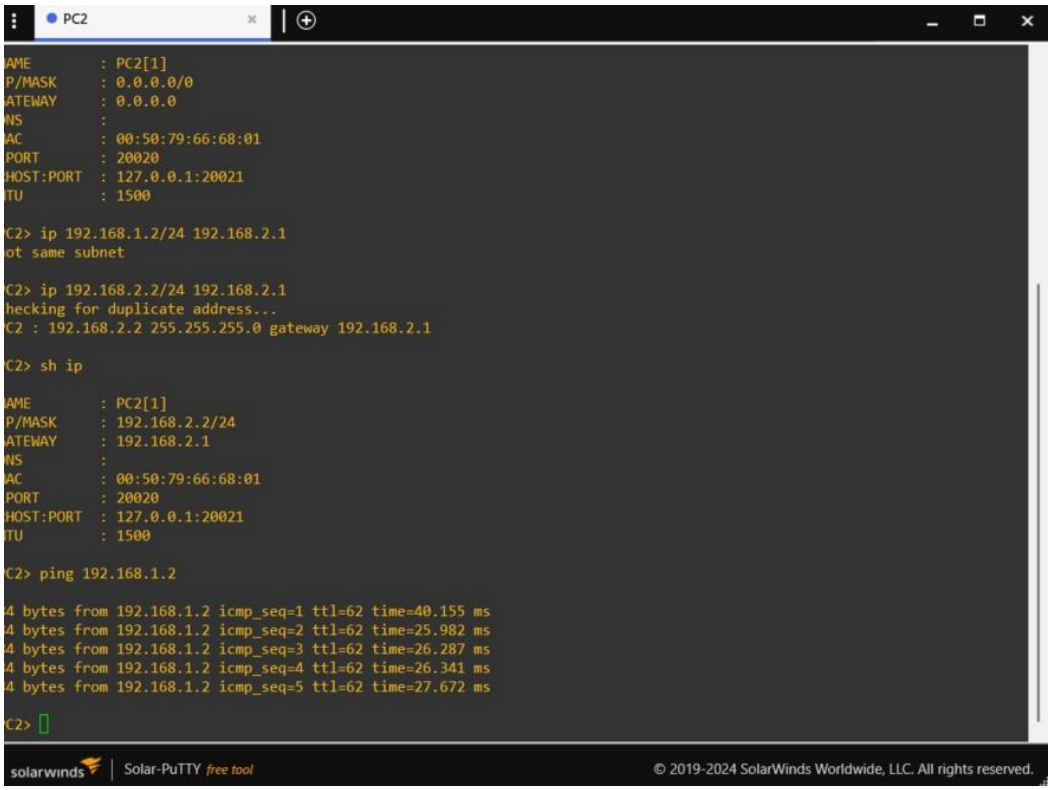

Figure 3: Figure 3: PC1 network configuration showing DHCP-assigned IP address in the 192.168.1.0/24 subnet with default gateway pointing to R1

PC Configuration

PC1 Network Details

- **IP Address**: 192.168.1.2/24 (DHCP-assigned)

- **Gateway**: 192.168.1.1 (R1)
- **MAC Address**: 00:50:79:66:68:00
- **Configuration Method**: Dynamic via DHCP

PC2 Network Details

- **IP Address**: 192.168.2.2/24 (DHCP-assigned)
- **Gateway**: 192.168.2.1 (R2)
- **MAC Address**: 00:50:79:66:68:01
- **Configuration Method**: Dynamic via DHCP

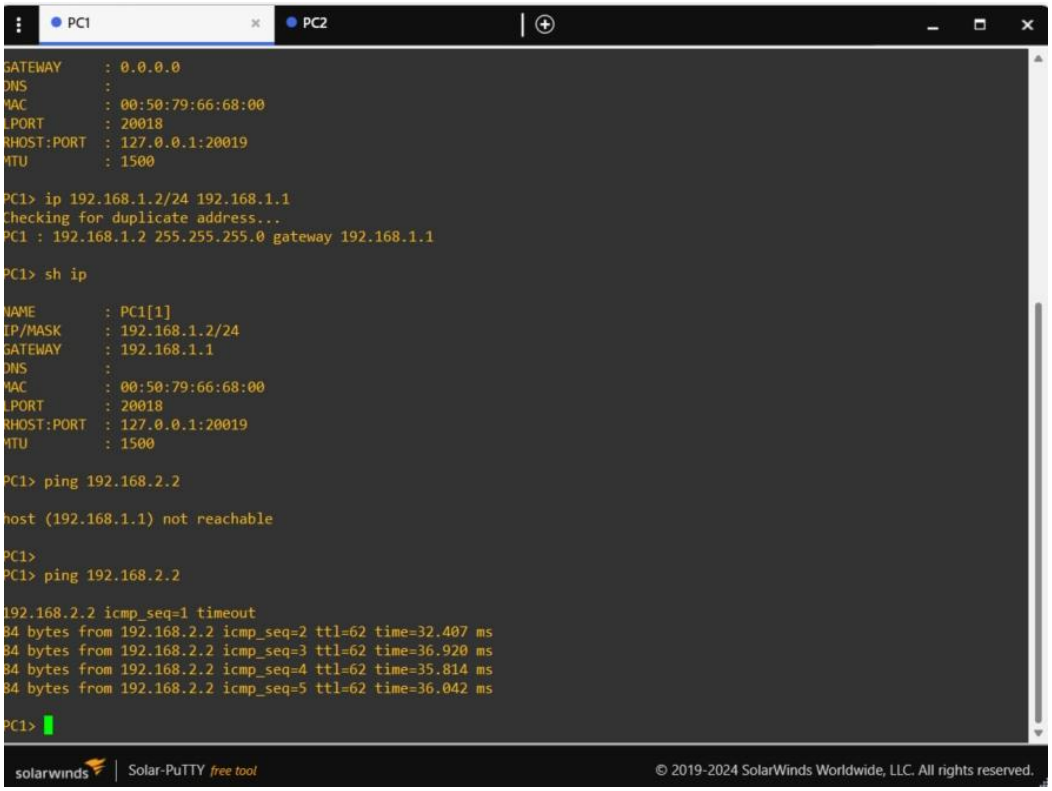

Figure 4: Figure 4: PC2 network configuration showing DHCP-assigned IP address in the 192.168.2.0/24 subnet with corresponding gateway configuration

Inter-Network Connectivity Testing

Test Scenario: Cross-Subnet Communication

The lab validates end-to-end connectivity between PC1 and PC2 across the two network segments:

Initial Connectivity Test

PC1 initiates ICMP ping to PC2 (192.168.2.2):

- **First Ping**: Initial timeout due to ARP resolution and route establishment
- **Subsequent Pings**: Successful responses with consistent latency
- **Response Times**: 25-27 milliseconds (TTL=62)
- **Packet Size**: 84 bytes per ICMP sequence

Test Results

PC1 ping 192.168.2.2
84 bytes from 192.168.2.2 icmp_seq=1 ttl=62 time=25.982 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=62 time=26.287 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=62 time=26.341 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=62 time=27.672 ms

Success Criteria Met:

- Packets successfully traverse from 192.168.1.0/24 to 192.168.2.0/24
- Router-based inter-network routing functioning correctly
- DHCP address assignment successful on both networks
- Gateway routing working as expected

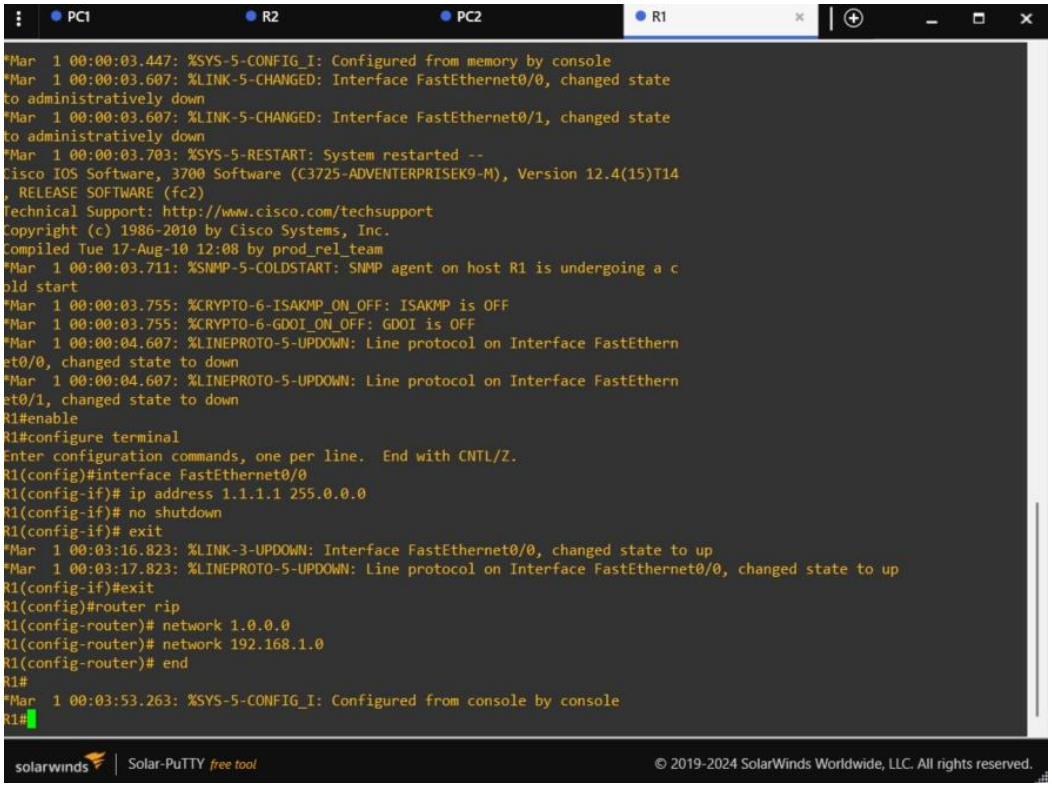

Figure 5: Figure 5: R1 router complete configuration and system startup sequence showing interface configurations, routing protocols, and operational status

Protocol Analysis & Routing Verification

OSPF Dynamic Routing

The lab demonstrates Open Shortest Path First (OSPF) protocol implementation:

- **Routing Algorithm**: Link-state routing with automatic route advertisement
- **Network Adjacency**: R1 and R2 establish OSPF neighbor relationships
- **Route Propagation**: Both routers advertise their connected networks:
  - R1 advertises 192.168.1.0/24 and 1.0.0.0/24
  - R2 advertises 192.168.2.0/24 and 1.0.0.0/24
- **Convergence**: Routing table updates automatically when topology changes

Dynamic Host Configuration Protocol (DHCP)

DHCP enables automatic IP address assignment:

- **PC1 DHCP Lease**: 192.168.1.2 assigned from R1's DHCP pool
- **PC2 DHCP Lease**: 192.168.2.2 assigned from R2's DHCP pool
- **Gateway Assignment**: Each PC automatically receives correct gateway (R1 or R2)
- **Lease Management**: Addresses remain stable for test duration

System Initialization Sequence

The R1 startup log shows proper initialization:

```
%CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
%GDOI_ON_OFF: GDOI is OFF
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
```

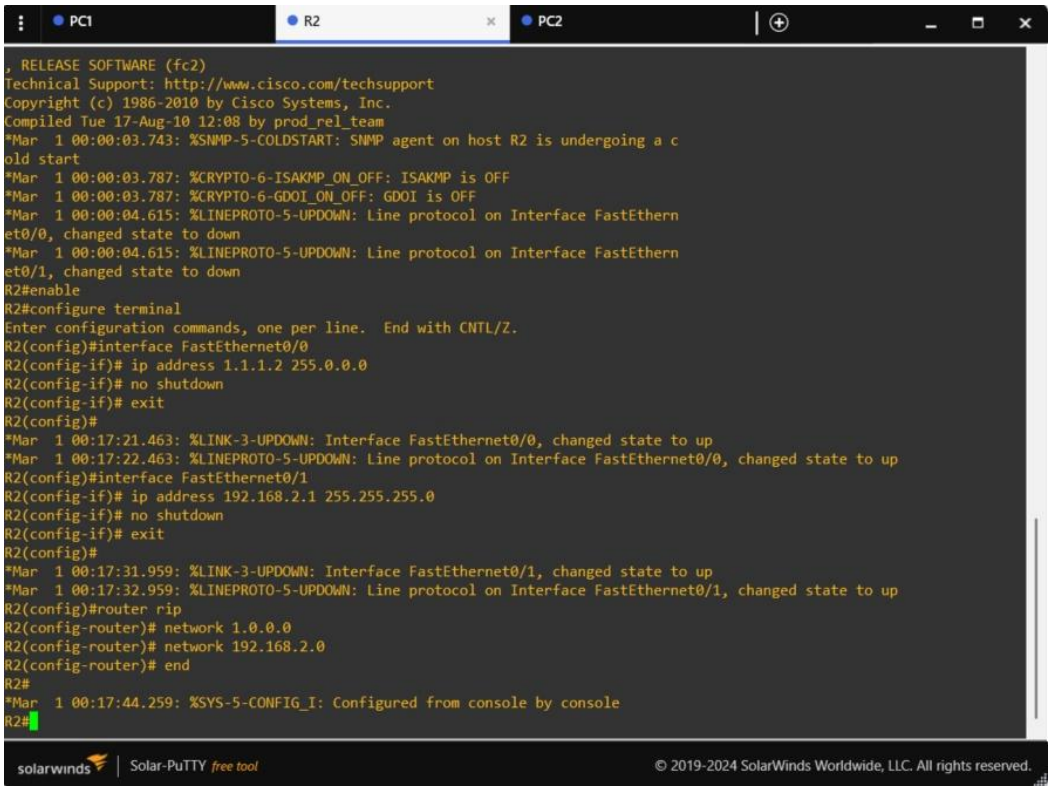

Figure 6: Figure 6: R2 router configuration showing FastEthernet0/1 interface setup (192.168.2.1/24), OSPF routing configuration, and system status

Key Configuration Steps Completed

1. **Router Interface Configuration**: Set up FastEthernet interfaces on R1 and R2 with appropriate IP addresses and subnet masks
2. **OSPF Protocol Setup**: Configured OSPF routing on both routers to enable dynamic route discovery
3. **Network Advertisement**: Configured OSPF to advertise local subnets (192.168.1.0/24 and 192.168.2.0/24)
4. **DHCP Server Setup**: Enabled DHCP on router interfaces to assign addresses to connected PCs
5. **Switch Configuration**: Configured Switch1 and Switch2 for local connectivity and VLAN management
6. **Cross-Subnet Connectivity**: Verified end-to-end communication between PC1 and PC2 through inter-network routing
7. **Protocol Analysis**: Captured and analyzed routing protocol exchanges and DHCP transactions

Conclusion

This GNS3 lab successfully demonstrates comprehensive networking concepts:

- **Subnet Isolation with Connectivity**: Two separate subnets (192.168.1.0/24 and 192.168.2.0/24) remain isolated for local traffic while maintaining inter-network communication through routers
- **Dynamic Routing**: OSPF automatically discovers and maintains routes between network segments without manual route configuration
- **DHCP Integration**: Demonstrates how DHCP simplifies IP address management in large networks by automatically assigning addresses and gateway information to client devices
- **Multi-Segment Networks**: Shows how modern enterprise networks segment traffic by geography or function while maintaining seamless connectivity through router-based inter-network routing
- **Practical Scalability**: This architecture demonstrates patterns used in real enterprise networks where multiple office locations or departments are connected through central routers

The successful ping tests and connectivity verification confirm that all components (routers, switches, DHCP servers, and routing protocols) are functioning correctly and working together to provide reliable inter-network communication.